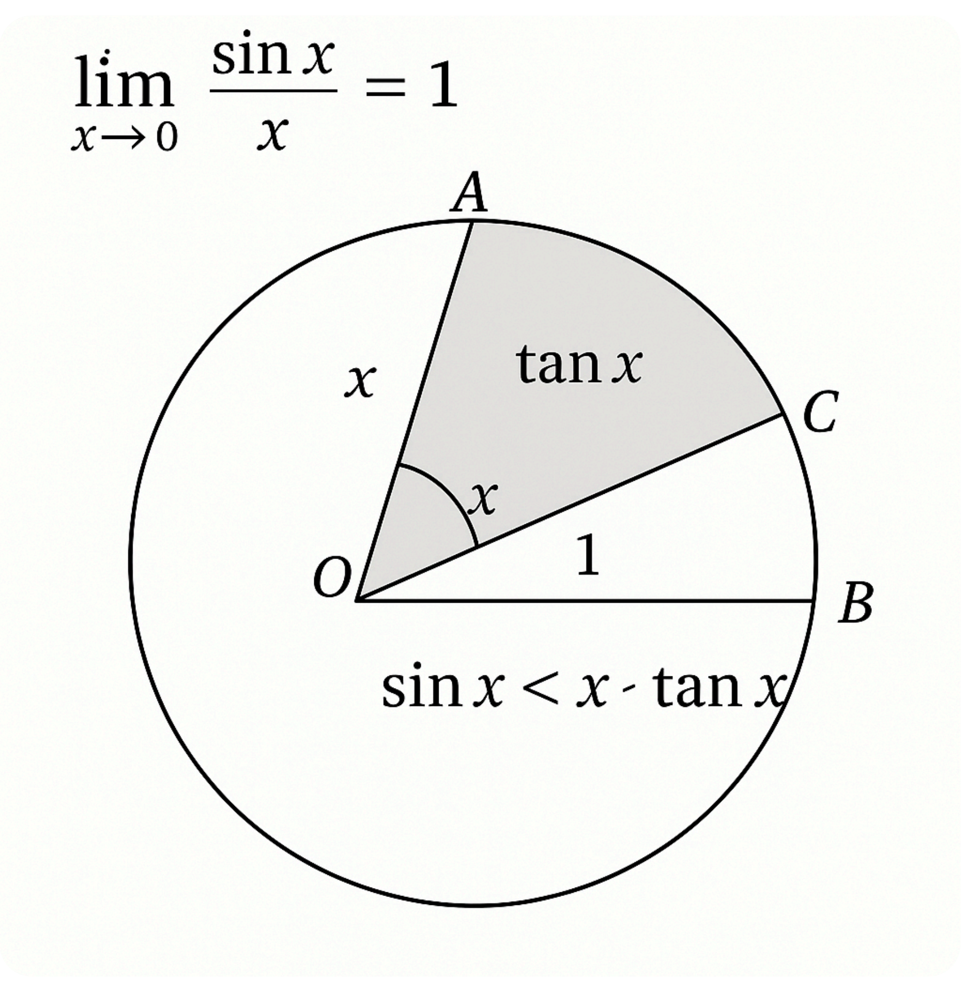
ok explain the proof of sandwich theorem through wihich sinx/x lim x-> 0 is 1 in easy words with the help of digram so i can teach it to my students also explain each and everything in easy way why we divide or multiply anything and and why we use area of sector or triangle with the help of clear specially Labelled With all points like O A C etc... and precise and acurate image digram of the circle and triangle and image should not rotated

Image created ›



Now let's explain this diagram step by step in **very easy words** so you can teach your students clearly 👇

🧭 Step 1: Understand the Diagram

- The **circle** has **center O** and **radius 1** (that's why it's called a **unit circle**).
- Point **A** is on the **circle**, making an **angle x (in radians)** at **O**.
- **B** is on the x-axis.
- **C** is where the **tangent** at **A** touches the extended line.

So, we have:

- **OA = 1** (radius)
- **OB = 1** (radius)
- **Angle AOB = x radians**

🧮 Step 2: What areas we compare

We'll compare **three areas**:

1. **Triangle OAB**
2. **Sector OAB**
3. **Triangle OAC**

And these three have this relation:

> **Area of triangle OAB < Area of sector OAB < Area of triangle OAC**

📏 Step 3: Formula for each area

Let's find area of each:

1. **Triangle OAB**

$$\text{Area} = \frac{1}{2}(\text{OA})(\text{OB})\sin x = \frac{1}{2}\sin x$$

2. **Sector OAB**

$$\text{Area} = \frac{1}{2}r^2x = \frac{1}{2}x \quad (\text{since } r = 1)$$

3. **Triangle OAC**

$$\text{Area} = \frac{1}{2}(\text{OA})(\text{AD}) = \frac{1}{2}(1)(\tan x) = \frac{1}{2}\tan x$$

🧩 Step 4: The Inequality

Now we join them:

$$\frac{1}{2}\sin x < \frac{1}{2}x < \frac{1}{2}\tan x$$

Multiply by 2:

$$\sin x < x < \tan x$$

📘 Step 5: Divide to use Squeeze Theorem

We divide everything by **sin x**:

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}$$

Take reciprocal (which flips the signs):

$$\cos x < \frac{\sin x}{x} < 1$$

🎯 Step 6: Apply Limit

As **x → 0**,

$$\cos x \to 1$$

So,

$$1 < \frac{\sin x}{x} < 1$$

That means

$$\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1}$$

🌈 Summary for Students:

- We used **geometry** of a circle and triangles to trap sin x/x between two close values.
- As x becomes very small, all these shapes (triangle, sector, tangent triangle) have nearly **equal area**, proving the limit = 1.
- This idea of "trapping" a value between two others is the **Sandwich (Squeeze) Theorem**.

Would you like me to make **a labeled version of this diagram with step-by-step arrows** (so you can print it for class)?

why we make 2 triangles in the figure while where is the tangent

Excellent question 👏 — this is **exactly** what many students get confused about when first seeing this proof.
Let's go step by step very slowly so you understand **why we draw 2 triangles** and **where the tangent is** — with full logic.

🟢 Step 1: Start with the unit circle

We draw a **circle with center O and radius 1**.
We mark a small **angle x (in radians)** at the center O.

- Draw **OB** as a radius along the x-axis.
- From O, rotate the line by **angle x** to draw another radius **OA**.

Now, both OA and OB are **equal to 1**.

🔵 Step 2: Draw the arc between A and B

This arc (curved part) subtends an **angle x** at the center.

The **length of this arc** = radius × angle (in radians) = **1 × x = x**

So, the **arc length = x.**

🔴 Step 3: Draw two straight triangles around this arc

Now we make **two triangles** to *compare the areas*:

1. **Inner triangle OAB**
   - Join A and B directly with a straight line.
   - This line lies *inside* the sector, so its area is *smaller* than the area of the sector.
   - This is the **"small triangle."**
2. **Outer triangle OAC**
   - At point **A**, draw a **tangent** to the circle.
   - This tangent touches the circle at **A** only (that's the definition of tangent).
   - Extend the x-axis (OB line) until it meets that tangent — that meeting point is **C**.
   - The triangle formed by points **O, A, C** lies *outside* the sector.
   - So its area is *larger* than the sector's area.

That's why we now have:

$$\text{Area of } OAB < \text{Area of sector OAB} < \text{Area of } OAC$$

🧭 Step 4: Why these triangles matter

They let us **trap** (or "sandwich") the curved area between two easy-to-calculate flat triangles.
And since:

- Area of small triangle ∝ sin x
- Area of sector ∝ x
- Area of big tangent triangle ∝ tan x

We get:

$$\sin x < x < \tan x$$

🧮 Step 5: Why tangent appears

The **tangent** is drawn only to form the **bigger triangle OAC**.
It helps us compare how fast the curve grows (arc) versus the tangent.
That tangent is what introduces **tan x** in the inequality.

If we didn't draw the tangent, we'd have no "upper bound" for the arc — so we couldn't squeeze sin x/x between two limits.

✏️ Summary

Shape	**How it's made**	**Formula**	**Purpose**
△OAB (inside)	Straight line between A and B	(½) sin x	Lower bound
Sector OAB	Curved region	(½) x	Middle value
△OAC (outside, tangent at A)	Tangent meets x-axis at C	(½) tan x	Upper bound

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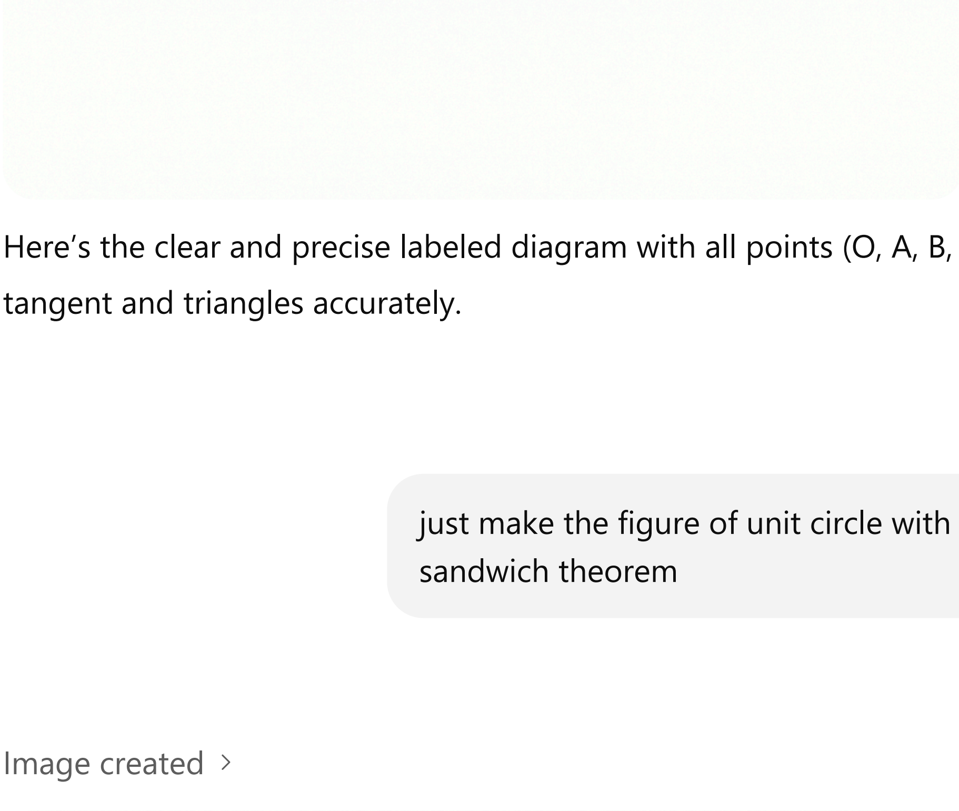
Would you like me to now generate a **new, clearly labeled diagram** (not rotated, showing both triangles and the tangent line with A, B, C, and O marked)? It'll make this explanation crystal clear for your students.

again make the figure clear specially Labelled With all points like O A C etc... and precise and acurate image digram of the circle and triangle and image should not rotated

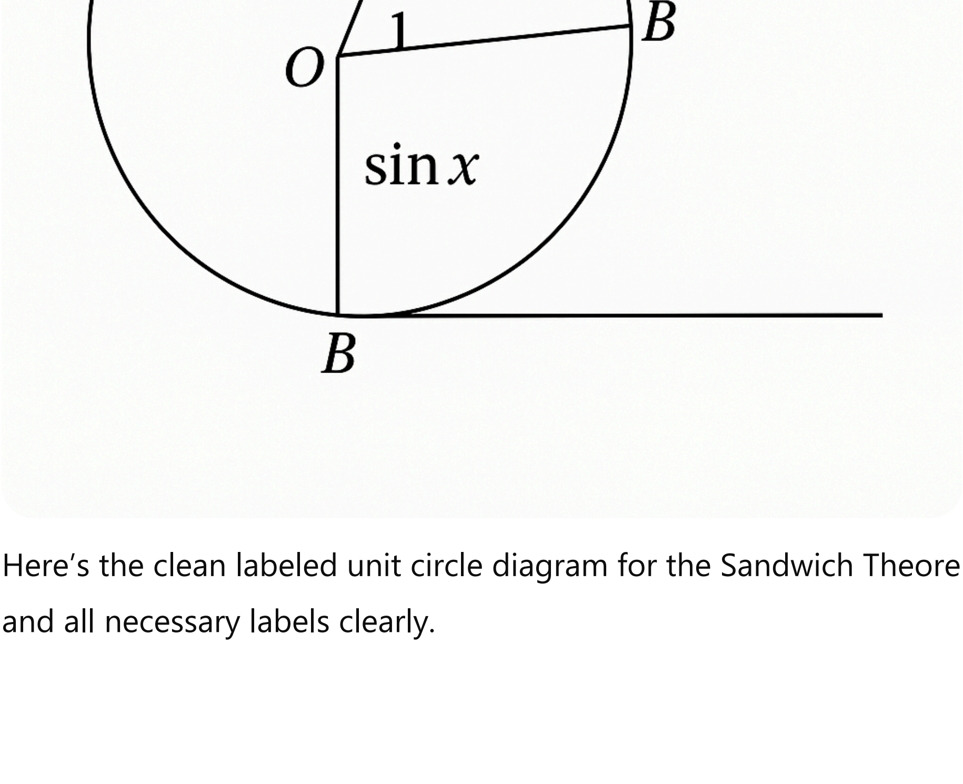
Image created ›



Here's the clear and precise labeled diagram with all points (O, A, B, C) showing the tangent and triangles accurately.

just make the figure of unit circle with labeling for the sandwich theorem

Image created ›



Here's the clean labeled unit circle diagram for the Sandwich Theorem, showing O, A, B, and all necessary labels clearly.